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Project Title:	Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres		
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Zip Code:	94720-1740	**Congressional District:**	9
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No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2026 per the PI (Ed., 9/16/25). NOTE: Start dates and end dates were updated per J. McQuillen/NASA-GRC. Original dates were 02/09/2022 - 02/28/2025 (Ed., 12/23/24).		
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COI Name (Institution):	Gollner, Michael (University of California, Berkeley)		
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Task Description:	NOTE: Continuation of NNX10AE01G, Material Ignition and Suppression in Space Exploration Atmospheres, with the same principal investigator, Prof. Carlos Fernandez-Pello. Central Objectives: A study is underway to conduct an experimental and theoretical research program on the effect of an external radiant flux on the ignition, subsequent flame spread, and fire suppression of a representative combustible material in space exploration atmospheres (SEA). The near-term objective of the research is to develop a ground based Material Ignition and Suppression Test (MIST) apparatus that reflects the environments that are expected in space-based facilities and is similar to the Solid Fuel Ignition and Extinction / Material Ignition and Suppression Test (SoFIE/MIST) apparatus that will be installed in the International Space Station (ISS). The ground based MIST apparatus has already been built and tested during the research program for NASA Grant #NNX10AE01G, and the apparatus is currently being used to obtain a flammability boundary, ignition, flame spread and extinction, of a polymeric material, specifically PMMA, which can be viewed as a baseline material for fire testing. As the microgravity data from the SoFIE/MIST experiments is available, it will be analyzed and compared with the normal gravity data to determine the effect of reduced gravity on the flammability of materials in SEA. Methods/techniques: The experimental apparatus consists of a small-scale rectangular flow duct that has a cylindrical fuel sample at the axis of the duct and has four radiant heaters placed in the corners of the duct that radiate the sample surface with a predetermined radiant flux. An electrical coil ignitor is placed at the trailing edge of the sample to ignite the sample and initiate flame spread. For SEA testing in normal gravity, the apparatus is placed inside a pressure chamber that resembles that of the Combustion Integrated Rack (CIR) in the ISS. An oxidizer gas, at SEA oxygen concentrations and pressures, flows at low velocity through the length of the duct. The fuel samples are rods of black PMMA of different diameters. The radiant heaters provide a radiant flux over the sample surface that increases its temperature and enhances the rate of flame spread. The ignitor provides the necessary energy to ignite the polymeric samples. The tests provide information about the effect of a radiant flux on the ignition, flame spread, and burning rate at a given oxygen concentration and ambient pressure. This will be followed by the reduction of the O2 concentration with N2 or CO2 until extinction occurs. The result will provide information about the suppression characteristics of inert gases in SEA and of the effect of SEA on the Limiting Oxygen Index (LOI) of the material when exposed to an external radiant flux. The experimental task will be complemented with a theoretical task that includes both analytical and numerical modeling. The analytical modeling uses simplified formulas of flame spread previously developed that incorporate the effect of the external radiant flux on the flame spread rate through the energy received by the fuel rod at a particular time. The numerical model will use a solid combustible pyrolysis numerical code developed in the Principal Investigator (PI) laboratory, together with a computational fluid dynamics (CFD) code simulating the reacting gas phase. The microgravity experiments conducted under the SoFIE project will be used to validate the theoretical model. It is the objective that the theoretical model will prevent the need to conduct extensive experimental testing of materials in microgravity. Perceived Significance: There is insufficient knowledge about fire behavior of materials in environments expected in space-based facilities, i.e., SEA (micro/partial-gravity, low velocity flow, reduced pressure, and elevated oxygen concentrations). Nor is there a testing methodology specifically developed to determine the fire hazards of materials under those conditions. It is the long-term objective of this project to provide additional information about the effect of SEA on the flammability (ignition, flame spread, and heat release rate) and suppression of materials, and guidance to interpret and extend current NASA testing procedures to environments expected in space-based facilities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project is intended to enhance the ability to predict ignition, burning (flame spread and mass burning), and extinction characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering.
	The research project has as objective to provide further information about the effect of SEA on the flammability of materials. Specifically, the evaluation of the ignition, flame spread, and extinction/limit oxygen concentration, of combustible materials to be potentially used in future manned spacecraft. For this purpose, an experimental, and theoretical research program is being conducted on the effect of space exploration atmospheres (SEA) on the flame spread characteristics of a representative combustible material (PMMA rods), and on their limits of oxygen concentration, and fire extinction conditions under an external heat flux. During this reporting period we have studied the opposed spread of flames over PMMA rods, and the subsequent downstream burning of the PMMA, as the oxygen in an enclosure is depleted by the burning PMMA. Emphasis is given to the differences in the limiting oxygen volume fraction for extinction (LOC) of flame spread and fuel regression in an enclosure at sub-atmospheric pressures, under normal and microgravity. More detailed information about the research conducted during this reporting period can be found in the cited publications. 1. Experiments on the opposed flame spread and fuel regression over PMMA rods in normal and microgravity The primary motivation of the work is to provide information about the extinction of solid fuels in atmospheres expected in futures spacecraft. Experiments with cylindrical black PMMA samples are conducted at sub-atmospheric pressures in a sealed combustion chamber on the Combustion Integrated Rack (CIR) of the International Space Station (ISS), and in a similar chamber in normal gravity. The oxygen concentration in the chamber is left to decrease naturally at constant pressure as the PMMA burns until extinction. The path to extinction of the opposed spread of flames and the fuel regression behind the flame is followed and their limiting oxygen volume fraction at extinction (LOC) recorded. It is found that there is a LOC at which the flame stops spreading, referred here as the limiting oxygen volume fraction for flame spread (LOC-FS), and another at lower oxygen concentration at which the regressing flame extinguishes (LOC-FR). It is also found that both LOCs decreases with ambient pressure and are significantly lower in microgravity than at Earth. The findings of this work show that current Earth-based flammability standards are not conservative enough and are insufficient for environments that future spacecraft are expected to operate in. Experimental Apparatus and methodology: Normal Gravity Experiments Experiments were carried out using a UCB dedicated Earth gravity apparatus, designed to investigate the flammability of solid combustion materials in controlled environmental conditions. The apparatus consists of a small-scale rectangular flow duct, with a 125 mm by 125 mm

cross section and 600 mm long. The sample is placed vertically at the center of the test section using an aluminum sample holder. Upstream, a duct section serves to straighten the flow before entering the test section. The apparatus is inserted in a 105 L pressure chamber. A fan located 15 cm below the test section provides an upward gas flow of 10 cm/s, similar to the air flow from the International Space Station's HVAC system. The samples used in the tests are black cast polymethyl methacrylate (PMMA) cylinders with dimensions of 12.7 mm,9.5 mm, and 6.4 mm in diameter and 55 mm in length. Experiments were carried out for different pressure conditions ranging from 100 kPa to 40 kPa.

Microgravity Experiments The microgravity tests were performed in the Material Ignition and Suppression Test (MIST) apparatus that was placed in the Solid Fuel Ignition and Extinction (SoFIE) testing facility. The SoFIE/MIST apparatus was contained in a 100 L combustion chamber in the Combustion Integrated Rack (CIR) of the Destiny module of the International Space Station. The SoFIE/MIST testing apparatus has similar characteristics and dimensions as the normal gravity MIST described above, although it is designed to minimize astronaut participation in the experiment by using robotic manipulation from the NASA Glen R.C. ground control center. . Experiment procedure During the tests, all inlets and outlets of the pressure chamber are closed, so that gradual oxygen depletion takes place in the chamber due to the PMMA's combustion. Once the chamber is sealed, a vacuum pump in normal gravity (1-g) or venting in micro-gravity is used to extract air from the chamber until the pressure at which the experiment would be performed is reached. The fan located upstream of the test section provides a 10 cm/s gas flow in the test section past the PMMA sample. The exhaust gases are recirculated in that chamber, ensuring a homogeneous mixing of gases in the test section. Ignition is induced with the help of an electrically heated nichrome wire at the top of the sample. The oxygen depletion data is gathered using a FDO2 optical oxygen gas sensor from PyroScience GmbH. The pressure in the chamber is monitored with a pressure transducer and kept constant by periodically extracting gas from the chamber when the pressure is above 2 kPa above the desired pressure. The flame spread, burning and extinction processes are video recorded with two cameras located outside the chamber. Subsequent analysis of the video images is performed using a FlameTracker processing application.

Experimental results

Flame Front and Flame Spread Rate The results of the tests conducted during this period show that both in 1-g and mg, as the oxygen concentration gradually decreases due to the combustion of the PMMA, the flame spread rate also decreases until it reaches an oxygen volume fraction where the flame no longer spreads. This oxygen threshold is defined here as the Limiting Oxygen Concentration (Volume Fraction) for Flame Spread (LOC-FS). It is also observed that after the LOC-FS is reached, as the oxygen is consumed further, there is a transition period where the flame leading edge moves gradually backward, in what appears to be a weak blow-off extinction. In normal gravity the flame eventually stops receding and attaches to the cone behind the pyrolysis front. As the PMMA surface keeps regressing, the cone angle becomes steeper, the oxygen concentration becomes smaller and the flame weaker, until it finally extinguishes. This oxygen threshold is defined here as the Limiting Oxygen Concentration (Volume Fraction) for Fuel Regression (LOC-FR). The LOC-FR threshold is the oxygen concentration normally recorded in the LOI apparatus since it reports the visual extinction of the flame. It is important to notice that the LOC-FR is significantly lower than the LOC-FS.

Task Progress:

It is also observed that reducing the chamber ambient pressure affects the flame's behavior acrossall measured parameters. Specifically, the LOC for the flame spread, and regression thresholds, are notably lower at reduced pressures. Correspondingly, the time required for the flame to reach these limits increases as the pressure decreases. Furthermore, the distance traveled by the flame during its active phases was consistently shorter at lower pressures. This reduction in flame travel distance at lower pressures suggests that the flame's ability to maintain sustained combustion over larger areas is compromised, likely due to the diminished oxygen availability and altered air density. Such observations are important for understanding the complex interplay between pressure, oxygen availability, and flame dynamics in confined spaces.

Flame spread rate vs oxygen concentration The measured flame spread rate (FSR) as a function of ambient oxygen concentration for normal gravity (1-g) and microgravity show that at a a given oxygen concentration, the FSR is systematically lower in 1-g than in microgravity across the ambient pressures tested, reflecting the enhanced convective losses imposed by buoyancy-driven flow. In microgravity, where buoyancy is absent, the flame maintains a higher spread rate for the same oxygen level and can spread at lower oxygen concentrations before propagation ceases. When comparing boundary cases, it is observed that at 40 kPa in microgravity, the FSR reaches zero at a slightly lower oxygen level than at 100 kPa in 1-g. This suggests that the absence of buoyancy compensates for the reduction in pressure, lowering the oxygen requirement for flame spread in microgravity relative to normal gravity.

Summary The results of these experiments show that the limiting oxygen concentration for opposed flame spread (LOC-FS) and for flame extinction during fuel regression, or mass burning, (LOC-FR) are different, with the former significantly higher than the later, in all the ambient conditions tested. The results show that at both gravitational conditions, reducing ambient pressures reduce the oxygen concentration limits for flame spread (LOC-FS) and fuel regression (LOC-FR), and decrease the time required to reach these limits. Furthermore, the effects are more pronounced under microgravity conditions, reflecting the sensitivity of flame behavior to ambient pressure at low gas velocities and in the absence of gravitational forces. The smaller LOC for flame spread and flame extinction during flame regression observed in micro-gravity are relevant because they show that PMMA is more flammable in microgravity than in normal gravity, at least for the current experimental conditions. This suggests that flammability standards are not conservative enough and are insufficient for environments that future spacecraft are expected to operate in.

A detailed description of the experimental result is presented in the annual summary report for the present grant period.

2 Modeling Extinction of Opposed Flame Spread over PMMA Rods

In this task, an assessment of the proposed model to determine the effect of the environmental variables on the LOC for thick PMMA sheets, and rods was conducted. The analysis is based on the condition that caused the flame spread rate, or the fuel regression rate, to become zero, i.e., no burning. The analysis predicts that for both flame spread and fuel regression, there are two oxygen volume fraction boundaries for extinction: a "heat transfer" boundary and a "chemical kinetics" boundary. The former is given by the conditions needed to heat the solid to its pyrolysis temperature, and the latter by the conditions needed for the gas phase reaction to be maintained. The combination of the two boundaries provides an overall boundary for the oxygen volume fraction for extinction as a function of material properties, gas flow velocity, ambient pressure, gravity level, and external heating.

The effect of the flow velocity on the LOC was particularly emphasized because it is the parameter with more data

available. The results obtained for the oxygen mole fraction (LOC) of thick PMMA sheets, and rods, as a function of the flow velocity, show that at very low flow velocities like those experienced in reduced gravity or microgravity, the extinction limit occurs at low flow velocities. At very low flow velocities in reduced gravity, the heat flux from the flame to the solid would be small because the flame standoff distance would be large, and the surface radiative heat losses can become critical in determining the LOC. This limit has been referred to as the radiation extinction limit. Furthermore, under these conditions the transport of the ambient oxygen to the flame would be reduced, which would decrease the flame temperature and in turn cause its extinction. This limit has been also referred to as the quenching limit. As the flow velocity is increase, the “heat transfer” oxygen mole fraction that leads to the flame extinction decreases (blue dotted line), this is due to the increased convective heat flux from the flame to the solid ahead of the flame. For fuel regression as the flow velocity increases the radiation heating from the flame to the fuel increases as the flame moves closer to the fuel surface. As the flow velocity increases further, the Damkohler number decreases, and the chemical kinetics boundary becomes dominant resulting in the LOC decreasing as the flow velocity increases. This limit is often referred to as the blow-off limit.

Summary An analytical theoretical model is presented to address solid fuel flammability, particularly extinction, starting from a fundamental definition of the flame spread and fuel regression processes. The model is developed to quantify the limiting oxygen mole fraction for extinction (LOC) considering the mechanisms affecting the solid fuel heating and the gas chemical kinetics controlling the spread of the flames over a thick solid. A discussion is introduced to address the influence of environmental variables such as flow velocity, ambient pressure, and external heating on the LOC. The developed model is compared with experimental data available in the literature and obtained in the above described experiments to further explore and demonstrate the influence of relevant parameters and elucidate the controlling factors in the extinction mechanisms. The developed model captures well the dependence observed in the experimental data on the flow velocity. This indicates that the proposed mechanisms leading to the extinction of a spreading flame over a solid surface and the subsequent fuel regression burning, and their analytical formulation, are correct and capable of predicting the LOC of a solid combustible material at least qualitatively.

Bibliography Type:	Description: (Last Updated: 02/15/2026)
Abstracts for Journals and Proceedings	C. Liveretou, J. Rivera, M. Bardy, C. Fernandez-Pello, M. Gollner, L. Etzenbach, M. Thomsen, S. L. Olson, and P. Ferkul "Flame extinction of thermally thick PMMA in Earth gravity versus microgravity: A comparison of Earth and on-orbit space experiments” " Journal of Fire Safety Science Journal of Fire Safety Science , Jun-2025
Abstracts for Journals and Proceedings	J. Rivera, C. Liveretou, C. Fernandez-Pello, M. J Gollner, X. Huang, S. Olson, P. Ferkul, and D. Stocker "“Opposed Flame Spread and Fuel Regression Over PMMA Rods in Normal and Micro Gravity: The Path to Extinction” " Fire Safety Journal, Fire Safety Journal, , Aug-2025
Abstracts for Journals and Proceedings	J. Rivera, C. Liveretou, C. Fernandez-Pello, M. J. Gollner, X. Huang, S. Olson, P. Ferkul "Opposed Flame Spread and Fuel Regression over PMMA rods in normal and microgravity: The Path to Extinction" 13th Mediterranean Combustion Symposium (13th MCS) 13th Mediterranean Combustion Symposium (13th MCS), Corfu, Greece, June 2025 , Jun-2025
Abstracts for Journals and Proceedings	M. Thomsen, C. Fernandez-Pello, D. L. Urban and S. L. Olson "Extinction of Opposed Flame Spread over a Thick Solid" 54th International Conference on Environmental Systems (ICES), Praga, Czech Republic , July (2025) 54th International Conference on Environmental Systems (ICES), Praga, Czech Republic , July (2025) , Jul-2025
Abstracts for Journals and Proceedings	J. Rivera, C. Liveretou, C. Fernandez-Pello, M. J. Gollner, D. Stocker, P. Ferkul, S. Olson "Olson “Opposed Flame Spread Limit over PMMA rods in normal and micro gravity" Fire Safety Journal, Fire Safety Journal, IAFSS special issue , Feb-2026